Detecção de Fraudes em Cartões de Crédito

```
import pandas as pd

url = "https://storage.googleapis.com/download.tensorflow.org/data/creditcard.csv"
df = pd.read_csv(url)
df.head()
```

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...	V21	V22
0	0.0	-1.359807	-0.072781	2.536347	1.378155	-0.338321	0.462388	0.239599	0.098698	0.363787	...	-0.018307	0.277838
1	0.0	1.191857	0.266151	0.166480	0.448154	0.060018	-0.082361	-0.078803	0.085102	-0.255425	...	-0.225775	-0.638672
2	1.0	-1.358354	-1.340163	1.773209	0.379780	-0.503198	1.800499	0.791461	0.247676	-1.514654	...	0.247998	0.771679
3	1.0	-0.966272	-0.185226	1.792993	-0.863291	-0.010309	1.247203	0.237609	0.377436	-1.387024	...	-0.108300	0.005274
4	2.0	-1.158233	0.877737	1.548718	0.403034	-0.407193	0.095921	0.592941	-0.270533	0.817739	...	-0.009431	0.798278

5 rows × 31 columns

Fechar

```
# PROBLEMA DE CLASSIFICAÇÃO DESBALANCEADA

  Fraudes são raras -> Modelo pode ignorar a classe 1
```

PROBLEMA DE CLASSIFICAÇÃO DESBALANCEADA

Fraudes são raras -> Modelo pode ignorar a classe 1

```
df["Class"].value_counts(normalize=True)
```

Class	proportion
0	0.998273
1	0.001727

dtype: float64

Feature Engineering

Criamos variáveis que ajudam o modelo

```
import numpy as np

df["Amount_log"] = np.log1p(df["Amount"])
```

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

df["Amount_scaled"] = scaler.fit_transform(df[["Amount"]])
```

```
from sklearn.model_selection import train_test_split

X = df.drop("Class", axis=1)
y = df["Class"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, stratify=y, test_size=0.3, random_state=42
)
```

LOGISTIC REGRESSION

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     85295
           1       0.84      0.66      0.73       148

    accuracy                           1.00     85443
   macro avg       0.92      0.83      0.87     85443
weighted avg       1.00      1.00      1.00     85443
```

Accuracy pode ser alta mesmo sem detectar fraudes

Por isso usamos:

- Recall (o mais importante)
- Precision
- F1-score

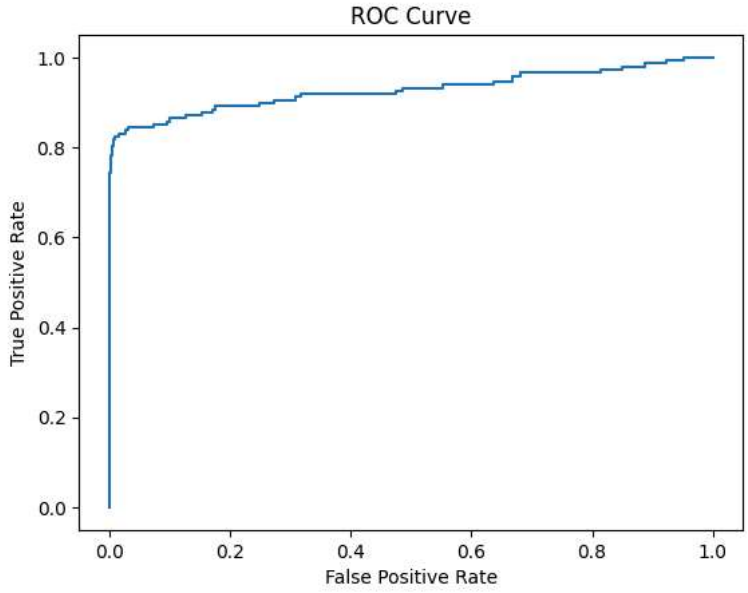
```
from sklearn.metrics import roc_curve, roc_auc_score
import matplotlib.pyplot as plt

y_probs = model.predict_proba(X_test)[:,1]

fpr, tpr, _ = roc_curve(y_test, y_probs)

plt.plot(fpr, tpr)
plt.title("ROC Curve")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.show()

print("AUC:", roc_auc_score(y_test, y_probs))
```

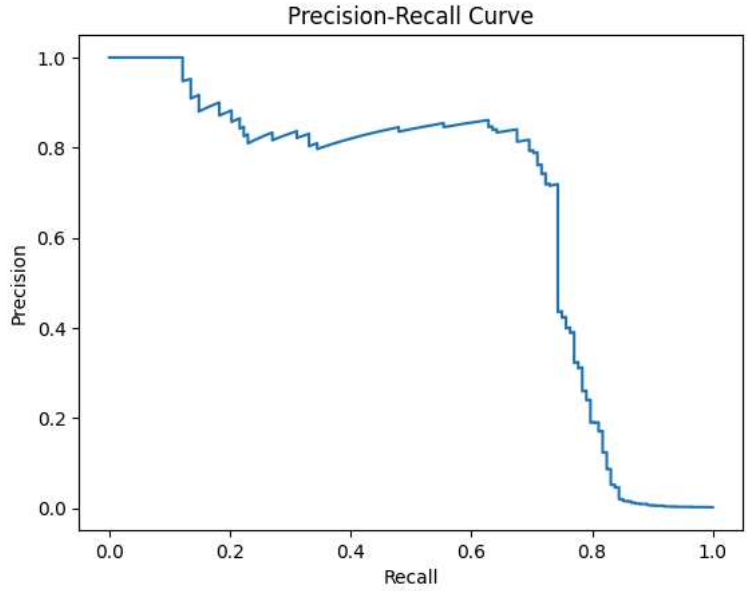



```
AUC: 0.9272042339543365
```

```
from sklearn.metrics import precision_recall_curve
import matplotlib.pyplot as plt

precision, recall, _ = precision_recall_curve(y_test, y_probs)

plt.plot(recall, precision)
plt.title("Precision-Recall Curve")
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.show()
```



BALANCEAMENTO DE DADOS

```
# Undersampling
fraudes = df[df["Class"] == 1]
normais = df[df["Class"] == 0].sample(len(fraudes), random_state=42)

df_under = pd.concat([fraudes, normais])
```

```
# Oversampling
from imblearn.over_sampling import SMOTE

smote = SMOTE()

X_res, y_res = smote.fit_resample(X, y)
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report

rf = RandomForestClassifier(
    n_estimators=50,
    max_depth=10,
    class_weight="balanced",
    n_jobs=-1,
    random_state=42
)

rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)

print(classification_report(y_test, y_pred_rf))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     85295
           1       0.84      0.76      0.79       148

    accuracy                           1.00     85443
   macro avg       0.92      0.88      0.90     85443
weighted avg       1.00      1.00      1.00     85443
```

```
from sklearn.pipeline import Pipeline

pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('model', LogisticRegression(max_iter=1000))
])

pipeline.fit(X_train, y_train)
y_pred = pipeline.predict(X_test)
```

```
threshold = 0.3
y_pred_custom = (y_probs > threshold).astype(int)

print(classification_report(y_test, y_pred_custom))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     85295
           1       0.79      0.71      0.75       148

    accuracy                           1.00     85443
   macro avg       0.89      0.85      0.87     85443
weighted avg       1.00      1.00      1.00     85443
```

MODELO MAIS AVANÇADO - XGBoost

XGBoost é um dos alogoritmos mais usados em competições e mercado

Ele é mais poderoso que Random Forest para muitos problemas

```
from xgboost import XGBClassifier

xgb = XGBClassifier(
    scale_pos_weight=10,  # ajuda com desbalanceamento
    use_label_encoder=False,
    eval_metric="logloss"
)

xgb.fit(X_train, y_train)

y_pred_xgb = xgb.predict(X_test)
```

```
/usr/local/lib/python3.12/dist-packages/xgboost/training.py:200: UserWarning: [07:06:49] WARNING: /__w/xgboost/xgboost/src/l
Parameters: { "use_label_encoder" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
```

```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred_xgb))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     85295
           1       0.93      0.78      0.85       148

    accuracy                           1.00     85443
   macro avg       0.96      0.89      0.92     85443
weighted avg       1.00      1.00      1.00     85443
```

```
Comece a programar ou gere código com IA.
```

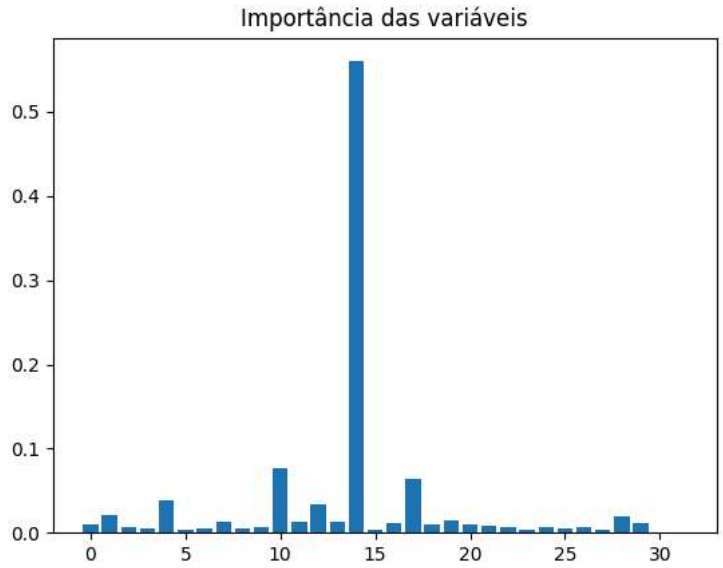
IMPORTÂNCIA DAS VARIÁVEIS

Ajuda entender quais variáveis influenciam mais o modelo

```
import matplotlib.pyplot as plt

# pegar importâncias do modelo XGBoost
importancias = xgb.feature_importances_

plt.bar(range(len(importancias)), importancias)
plt.title("Importância das variáveis")
plt.show()
```



AJUSTE DE HIPERPARÂMETROS

Testamos várias combinações para testar o modelo

```
from sklearn.model_selection import GridSearchCV
from xgboost import XGBClassifier

param_grid = {
    "max_depth": [3, 5],
    "n_estimators": [50, 100]
}

grid = GridSearchCV(
    XGBClassifier(eval_metric="logloss"),
    param_grid,
    scoring="recall",
    cv=3
)

grid.fit(X_train, y_train)

print("Melhor modelo:", grid.best_params_)
```

```
Melhor modelo: {'max_depth': 3, 'n_estimators': 100}
```

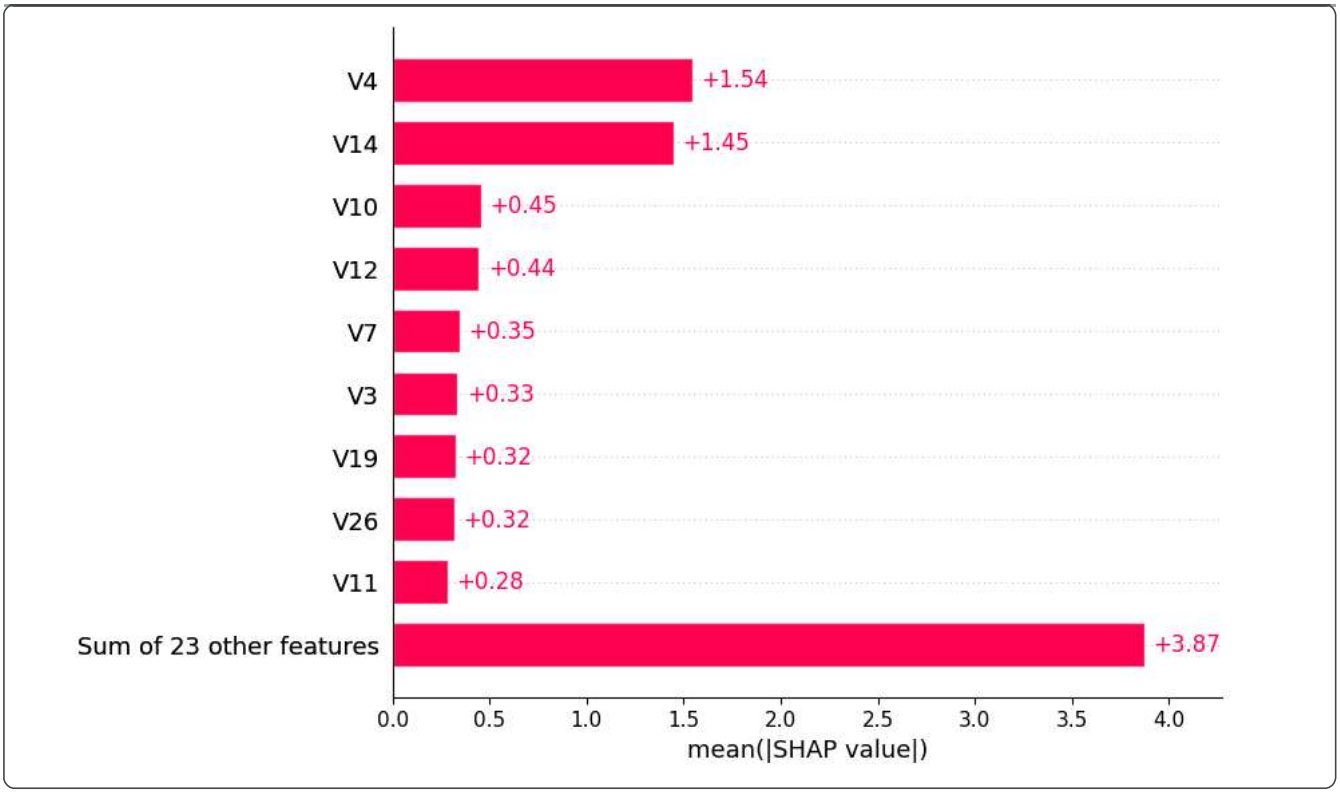
EXPLICABILIDADE (SHAP)

SHAP mostra como cada variável influencia a decisão do modelo

```
import shap

explainer = shap.Explainer(xgb)
shap_values = explainer(X_test[:100])

shap.plots.bar(shap_values)
```


V4
+1.54
V14
+1.45
V10
+0.45
V12
+0.44
V7
+0.35
V3
+0.33
V19
+0.32
V26
+0.32
V11
+0.28
Sum of 23 other features
+3.87
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
mean(|SHAP value|)